

Assignment module 2

Installation and Maintenance of Hardware and Its

Section 1: Multiple Choice

1. Which of the following precautions should be taken before working on computer hardware?

- a) Ensure the computer is plugged in to prevent electrostatic discharge.
- b) Wear an anti-static wrist strap to prevent damage from electrostatic discharge.
- c) Work on carpeted surfaces to prevent slipping.
- d) Use magnetic tools to handle components more easily.

Ans :- B

2. What is the purpose of thermal paste during CPU installation?

- a) To insulate the CPU from heat.
- b) To provide mechanical support for the CPU.
- c) To improve thermal conductivity between the CPU and the heat sink.
- d) To prevent the CPU from overheating

Ans :- C

3. Which tool is used to measure the output voltage of a power supply unit (PSU)?

- a) Multi meter
- b) Screwdriver
- c) Pliers
- d) Hex key

Ans:- A

4. Which component is responsible for storing BIOS settings, such as date and time, even when the computer is powered off?

- a) CMOS battery
- b) CPU
- c) RAM
- d) Hard drive

Ans :- A

5. True or False: When installing a new hard drive, it is essential to format it before use.

Ans :- True

6. True or False: A POST (Power-On Self-Test) error indicates a problem with the CPU.

Ans :- False

7. True or False: It is safe to remove a USB flash drive from a computer without ejecting it first.

Ans :- False

8. Describe the steps involved in installing a new graphics card in a desktop computer.

Ans :- steps are Like That.

- Power Off And Unplug The PC.
- Open The Cabinet
- if there is an old card and you want to remove and install new so remove it and install new card.
- secure the card
- connect the power cable
- close the cabinet
- turn on and install drivers

9. What is RAID, and what are some common RAID configurations?

Ans :- RAID is a data storage technology that combines multiple hard drives or SSDs to improve performance, redundancy, or both.

RAID 0 (Striping) – Improves performance by splitting data across multiple drives but offers no redundancy (if one drive fails, all data is lost).

RAID 1 (Mirroring) – Creates an exact copy of data on two drives, providing fault tolerance but no performance boost.

RAID 5 (Striping with Parity) – Requires at least 3 drives, offers redundancy by storing parity data, and allows one drive failure without data loss.

RAID 10 (1+0, Mirrored Striping) – A combination of RAID 1 and RAID 0, requiring at least 4 drives, offering both speed and redundancy.

10. Demonstrate how to replace a CPU fan in a desktop computer.

Ans:- Done in lab.

11. Discuss the importance of network documentation and provide examples of information that should be documented.

Ans :-

The Importance of Network Documentation

Network documentation is **crucial for managing, troubleshooting, and maintaining a network efficiently**. It provides IT professionals with a clear understanding of the network infrastructure, making it easier to diagnose issues, implement upgrades, and ensure security. Proper documentation helps minimize downtime, reduces misconfigurations, and enhances network scalability.

Benefits of Network Documentation:

1. **Easier Troubleshooting** – A well-documented network allows IT teams to quickly identify and resolve issues, reducing downtime.
2. **Efficient Network Management** – Clear documentation helps in managing devices, configurations, and access controls effectively.
3. **Improved Security** – Proper documentation includes security policies, firewall settings, and access permissions, reducing vulnerabilities.
4. **Simplified Onboarding and Training** – New IT staff can quickly understand the network setup, reducing training time.

5. **Regulatory Compliance** – Many industries require network documentation for compliance with standards like HIPAA, PCI-DSS, or ISO 27001.

Examples of Information That Should Be Documented:

1. **Network Topology Diagrams** – Visual representation of network structure, including devices and connections.
2. **IP Address Allocation** – List of assigned IP addresses, subnet masks, and DHCP configurations.
3. **Hardware Inventory** – Details of routers, switches, firewalls, and other networking devices.
4. **Configuration Settings** – Backup of router, switch, and firewall configurations.
5. **User Access Controls** – List of authorized users, access levels, and authentication methods.
6. **Security Policies** – Firewall rules, intrusion detection settings, and VPN configurations.
7. **Service Provider Information** – ISP details, bandwidth, and contact information for troubleshooting.